Integer and mixed-integer programming

2105623 · November 2, 2026

Choosing machines, opening production lines, and verifying decisions

Your three-hour study route

15 min review + 30 min integer geometry

35 min modeling + 10 min break

25 min search + 45 min lab + 20 min Exit Ticket

Recall the LP certificate

A feasible plan gives a candidate value.

A feasible dual solution bounds all primal values.

Matching values certify LP optimality.

Three kinds of decisions

Continuous: tonnes produced, $q \geq 0$.

Integer: number of machines, $n \in \{0,1,2,\ldots\}$.

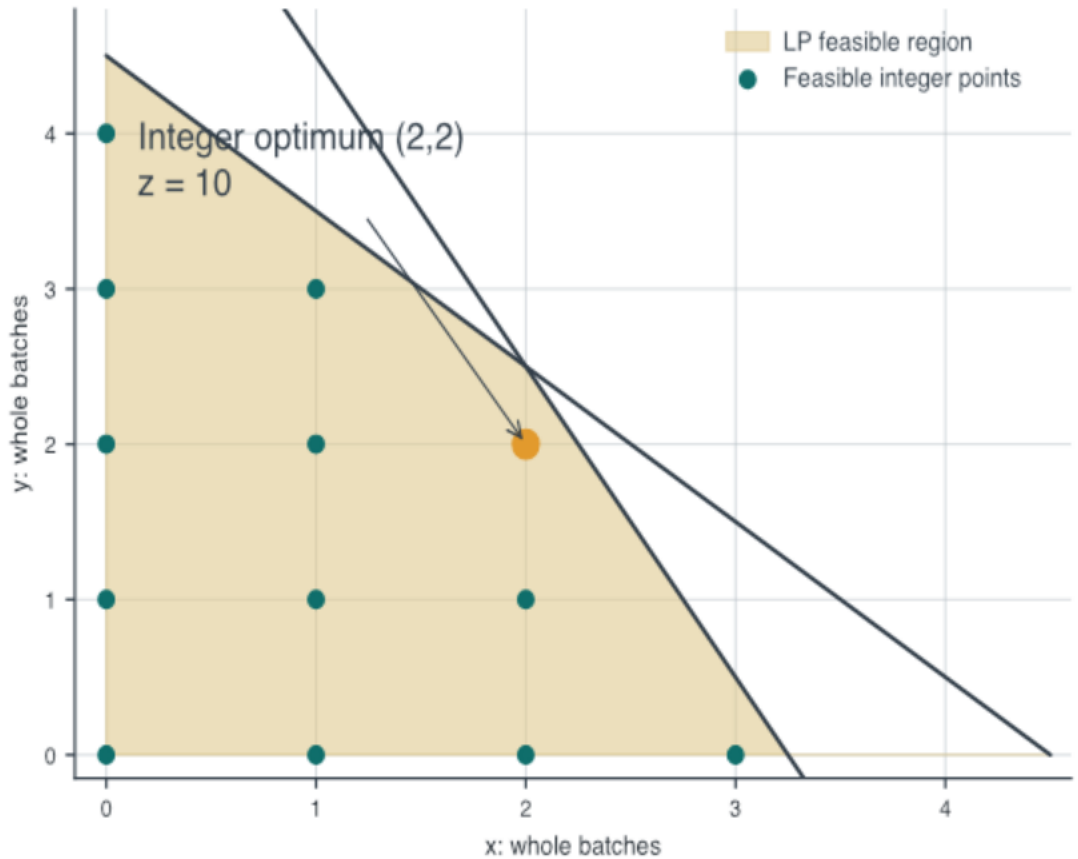
Binary: open a line, $y \in \{0,1\}$.

A small integer production problem

Maximize 3x + 2y.

$x + y \le 4.5$; $2x + y \le 6.5$.

x and y count batches: nonnegative integers.

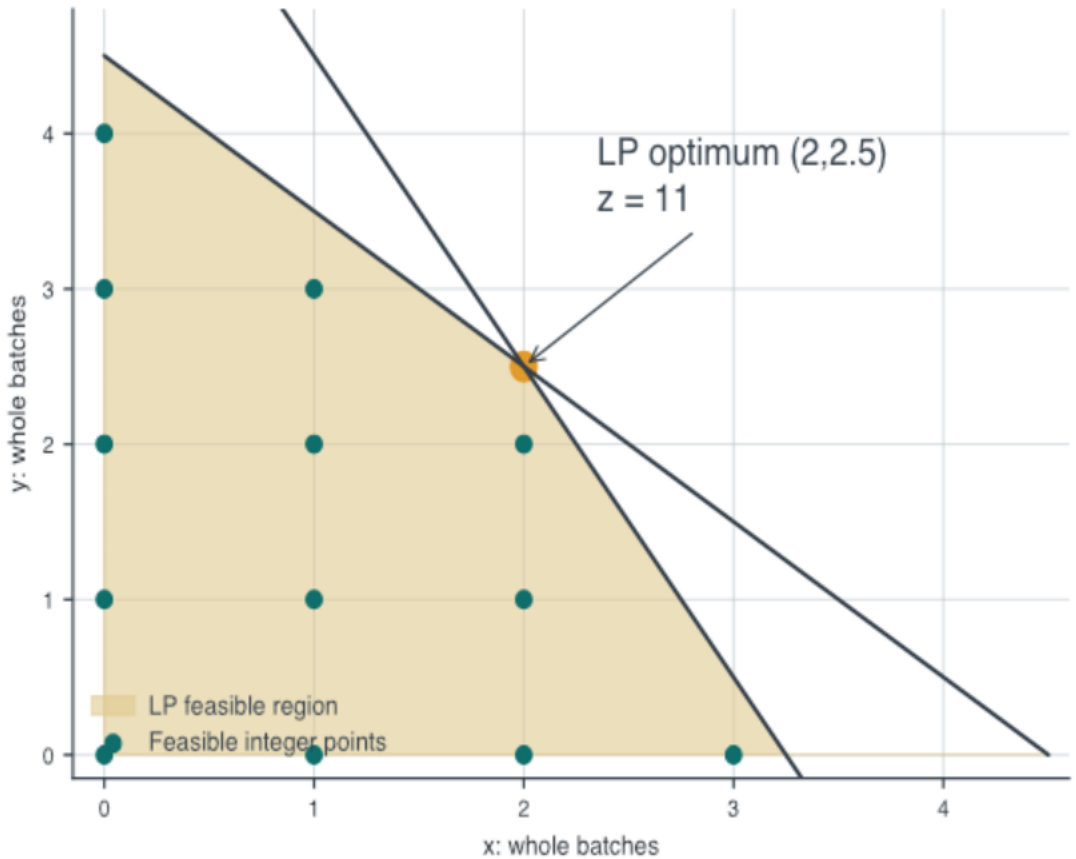


Solve the continuous relaxation

Intersection: x + y = 4.5 and 2x + y = 6.5.

Subtract: x = 2; then y = 2.5.

LP value = 3(2) + 2(2.5) = 11.

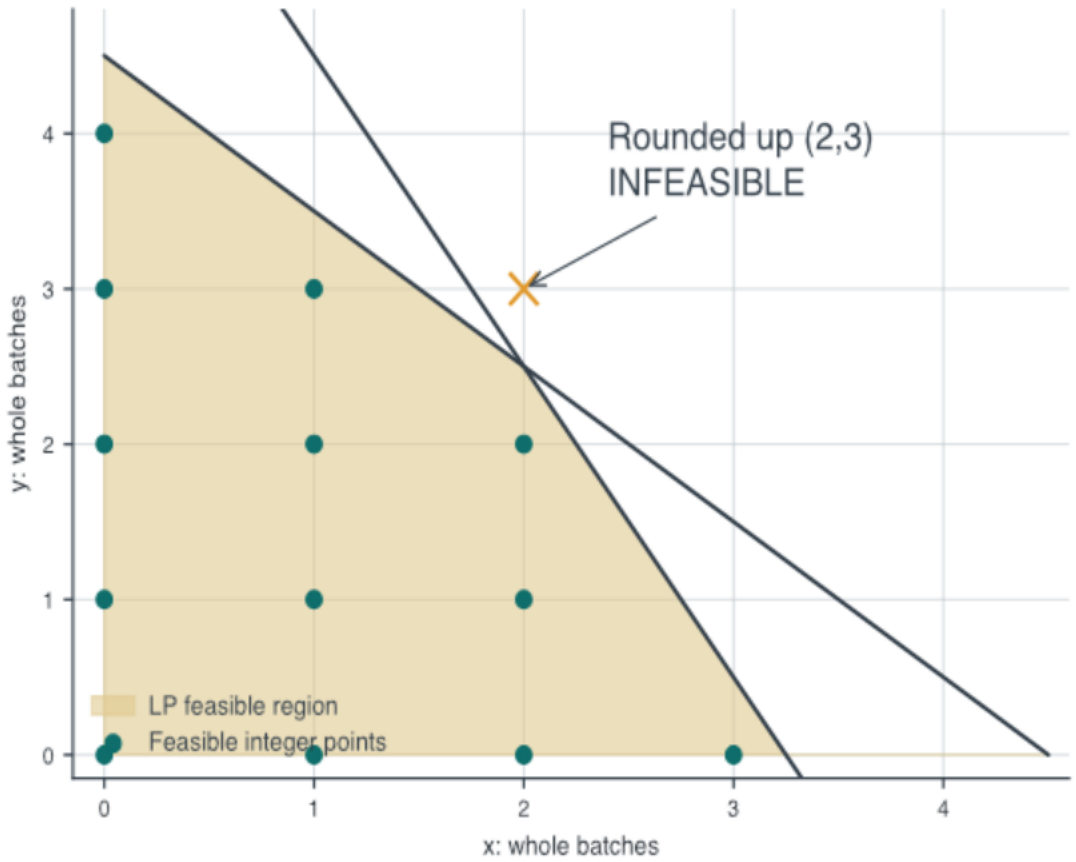


Rounding is not a solution method

Round upward to (2,3): resource use (5,7).

Both limits (4.5,6.5) are violated.

Rounding downward happens to work here; it needs proof.

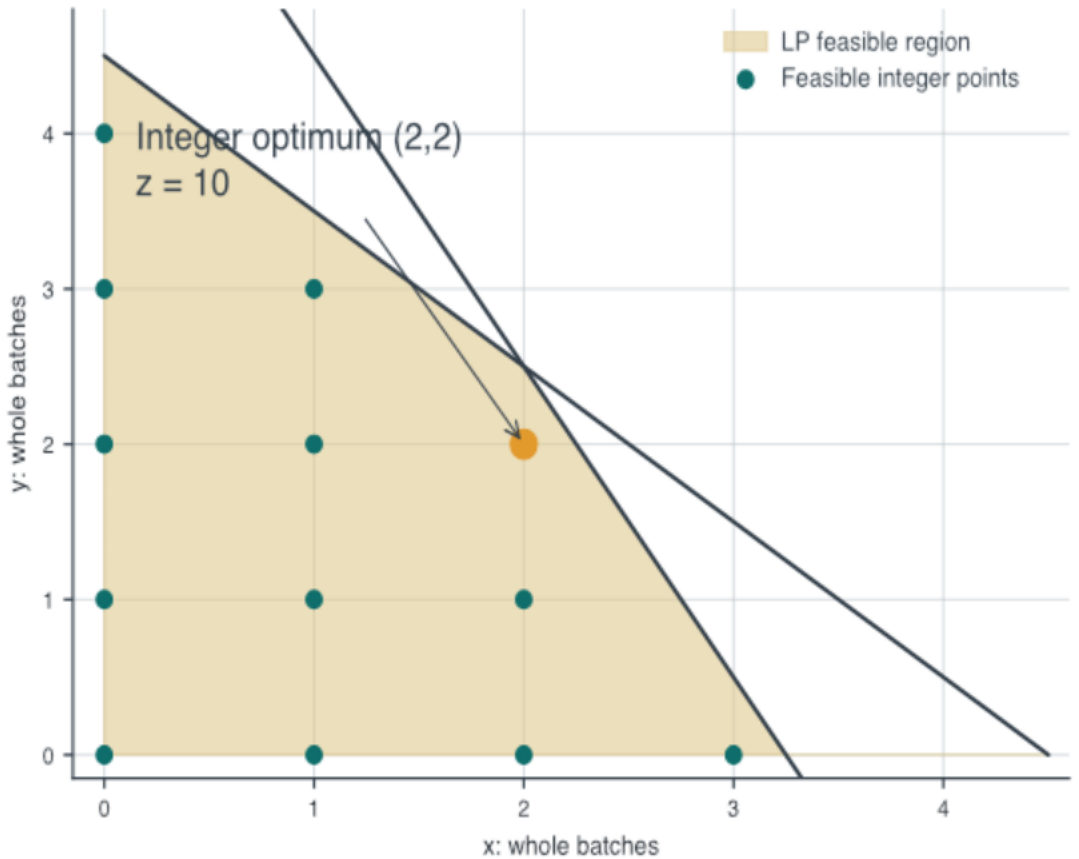


The integer optimum

Enumerate feasible integer points.

Best point: (x,y) = (2,2), value 10.

LP upper bound 11; integer value 10.



Bounds depend on objective direction

Maximization: feasible integer value ≤ optimum ≤ LP bound.

Minimization: LP bound ≤ optimum ≤ feasible integer value.

A bound is useful even when its solution is fractional.

Pause: check a proposed batch plan

For (x,y) = (3,1), calculate both resource uses.

Is the plan feasible? What is its objective?

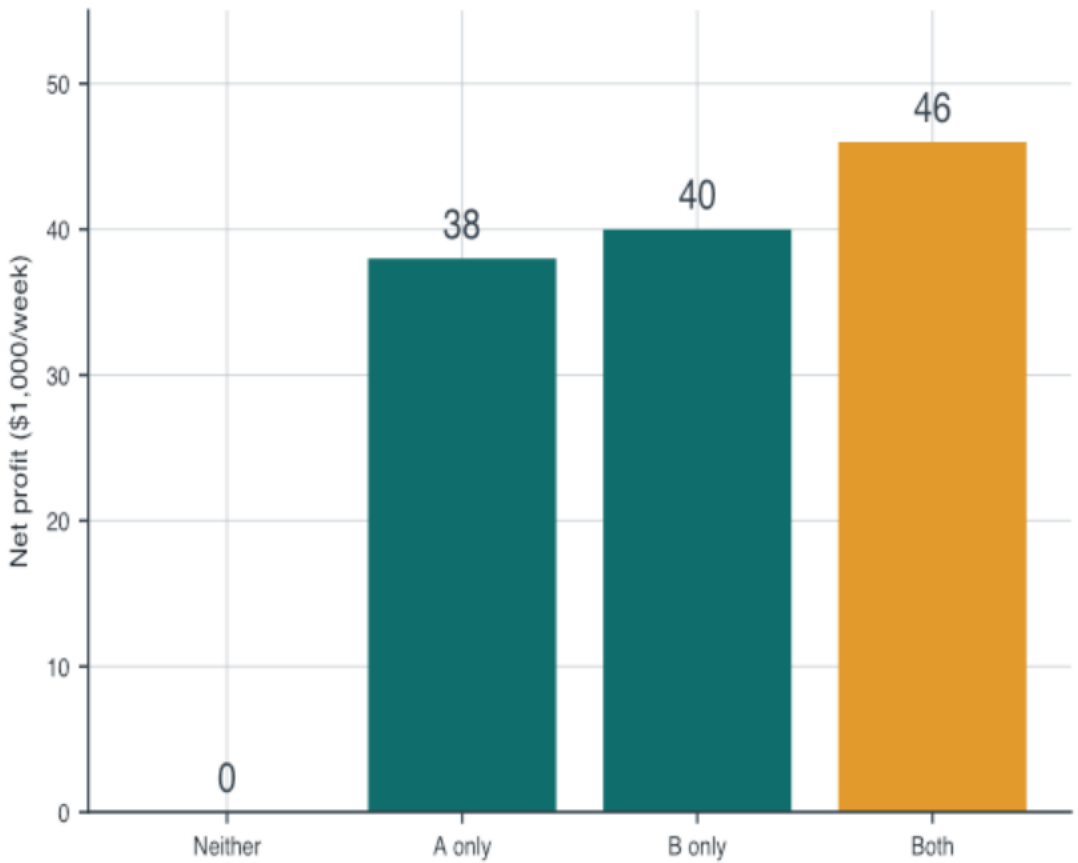
Explain why a high objective is not enough.

A plant with two optional lines

Illustrative weekly data; money in $1,000/week.

Line A: contribution 8/tonne; opening cost 10.

Line B: contribution 6/tonne; opening cost 8.

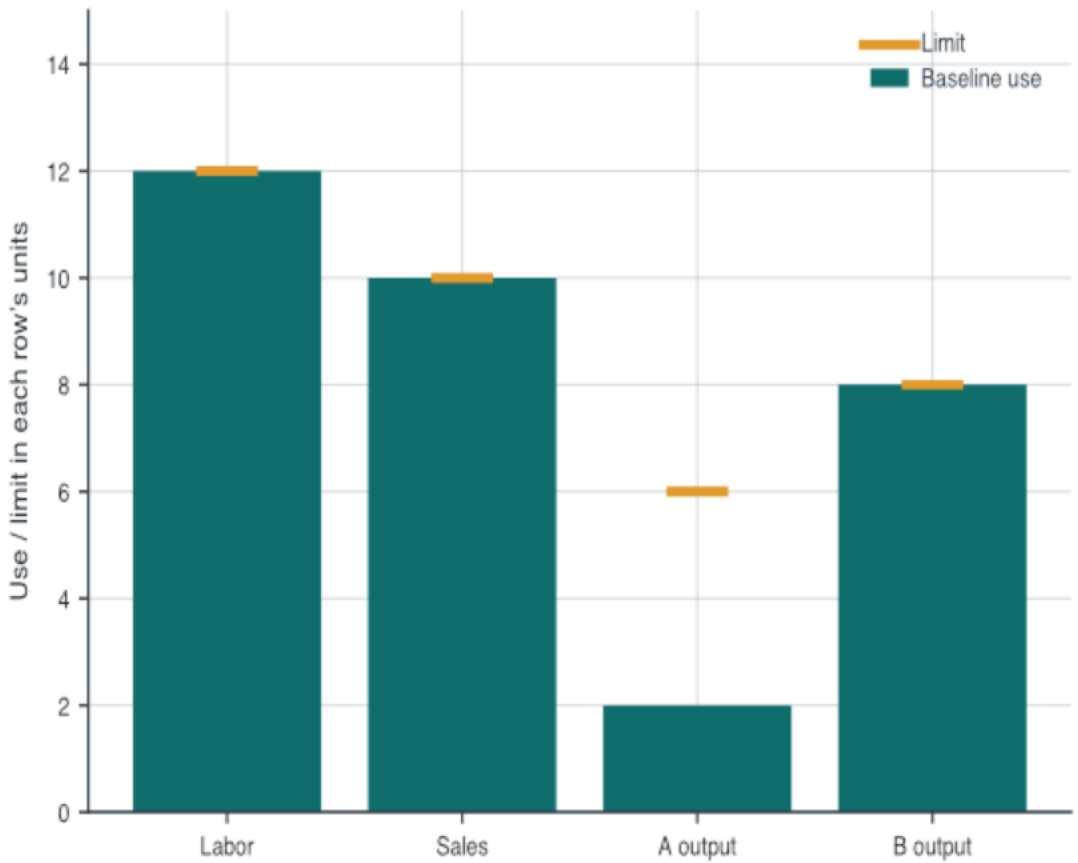


Capacity and resource data

A capacity 6 tonnes/week; B capacity 8.

Labor use: 2 units/tonne for A, 1 for B.

Labor ≤ 12 units/week; total sales ≤ 10 tonnes/week.



Separate production from opening

$a, b \geq 0$: continuous tonnes/week.

$yA, yB \in \{0,1\}$: whether each line is opened.

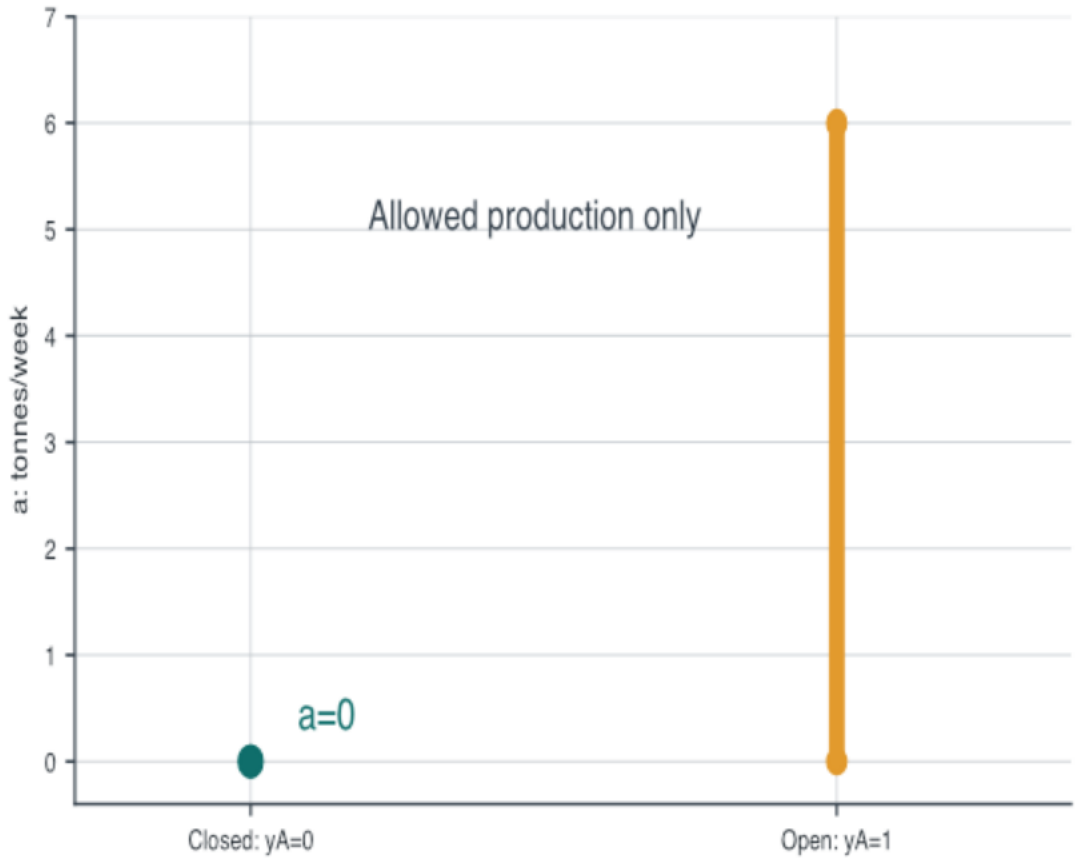
Opening permits production; it does not force full output.

Linking constraint: a ≤ 6yA

If yA = 0, then 0 ≤ a ≤ 0: a = 0.

If yA = 1, then 0 ≤ a ≤ 6.

Use b ≤ 8yB for line B.

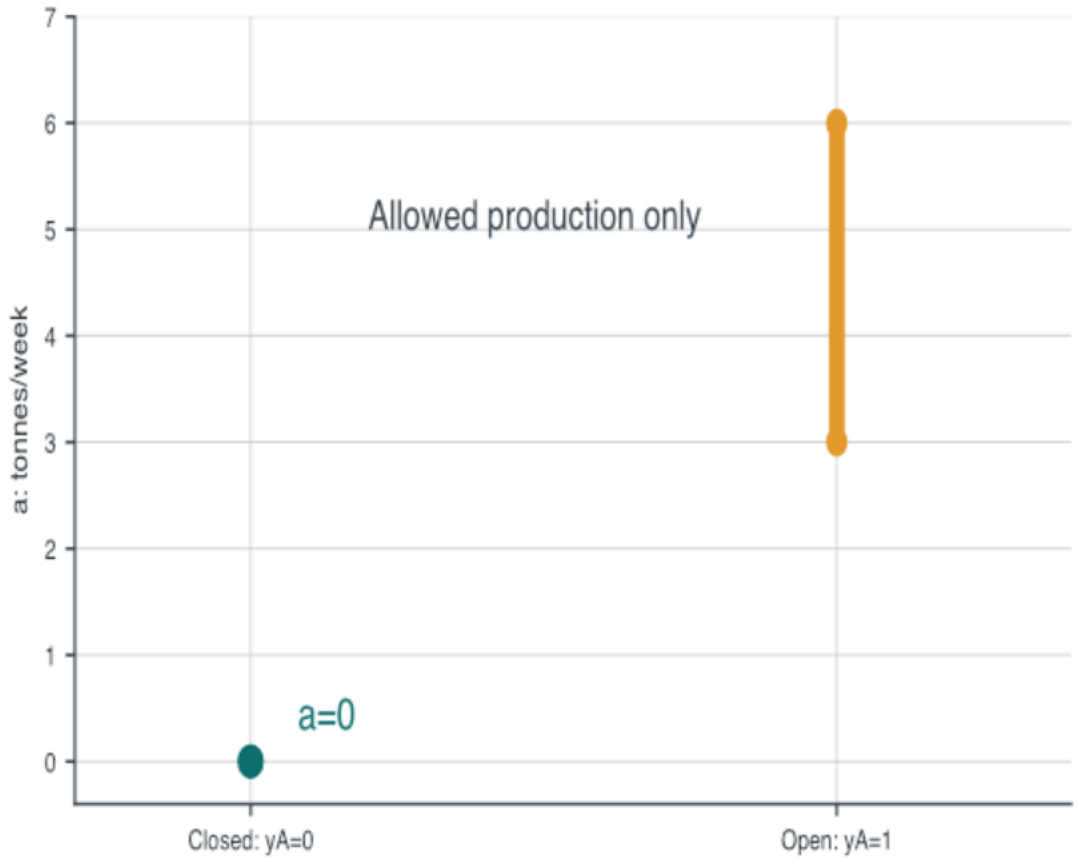


An optional minimum production level

If A opens, require at least 3 tonnes.

Add $a \geq 3y_A$ together with $a \leq 6y_A$.

Closed: $a=0$. Open: $3 \leq a \leq 6$.



Logic with binary variables

At most one line: $y_A + y_B \leq 1$.

Exactly one line: $y_A + y_B = 1$.

A requires B: $y_A \leq y_B$.

The complete baseline MILP

Maximize 8a + 6b − 10yA − 8yB.

2a + b ≤ 12; a + b ≤ 10.

a ≤ 6yA; b ≤ 8yB; a,b≥0; yA,yB binary.

$$\max\ 8a + 6b - 10y_A - 8y_B$$

$$2a + b \le 12, \quad a + b \le 10$$

$$a \le 6y_A, \quad b \le 8y_B$$

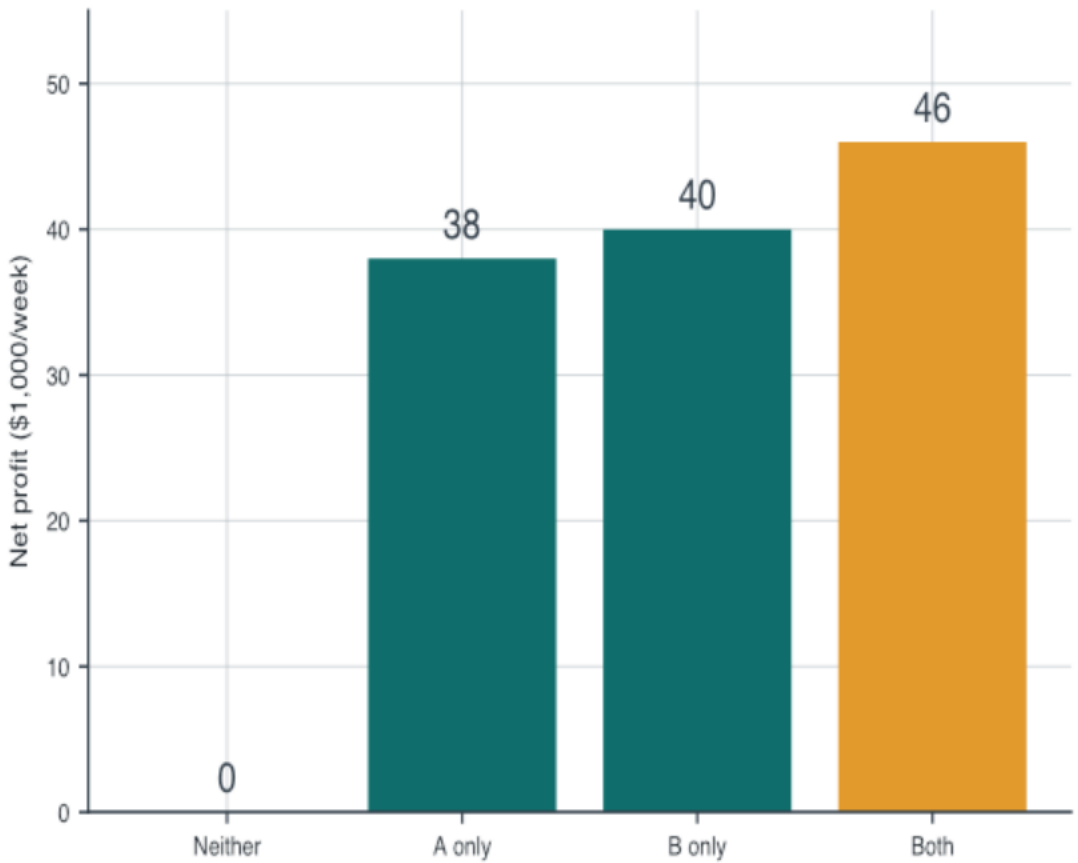
$$a, b \ge 0, \quad y_A, y_B \in \{0, 1\}$$

Verify by enumerating opening choices

Neither: profit 0. A only: a=6, profit 38.

B only: b=8, profit 40.

Both: a=2,b=8, profit 46.

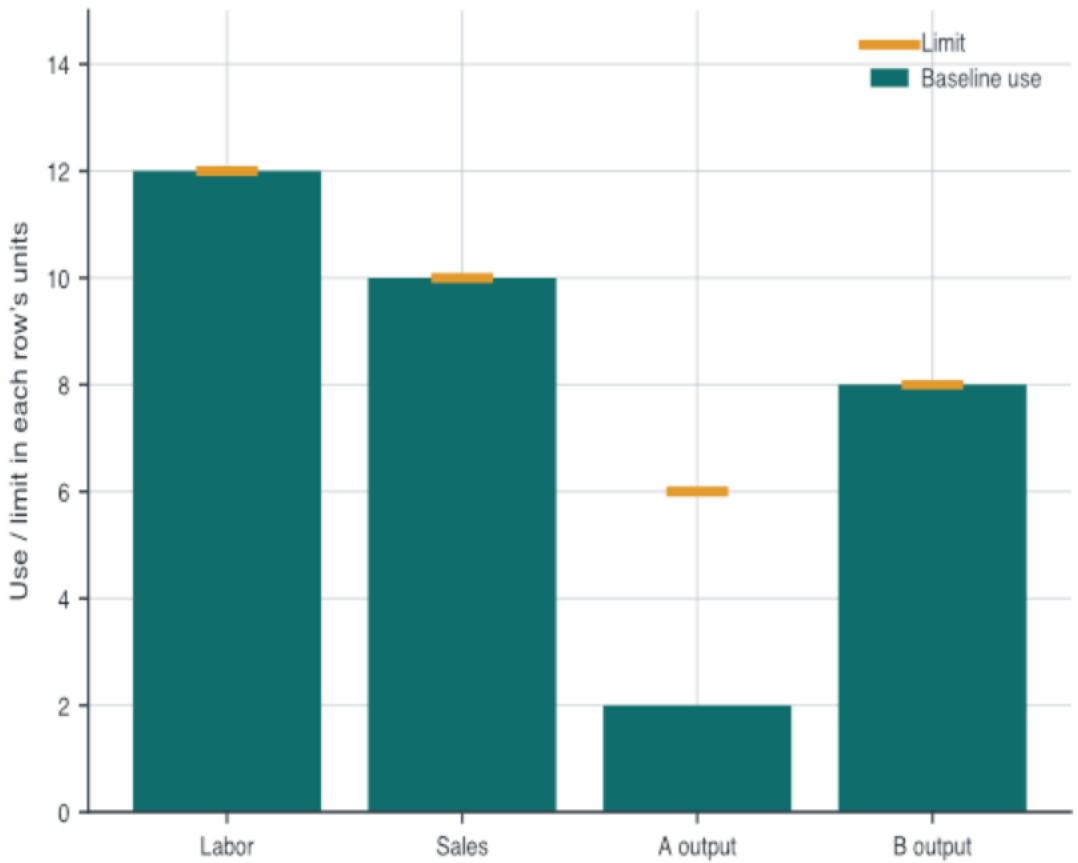


Audit the chosen production plan

a=2,b=8,yA=yB=1.

Labor: 2(2)+8=12. Sales: 2+8=10.

Profit: 16+48−10−8=46.

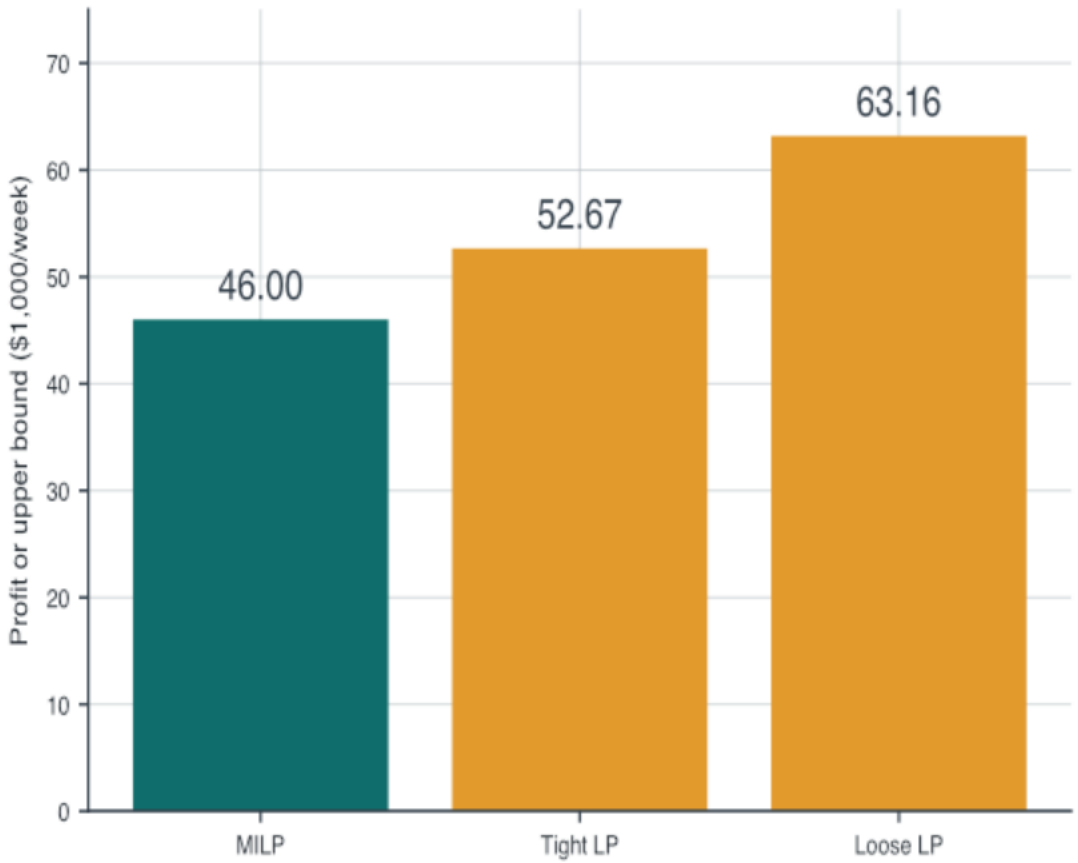


Choose a meaningful upper bound

Use the physical bound U=6 for $a \le U y_A$.

An unnecessarily large U weakens the LP relaxation.

Never choose U so small that valid plans are excluded.

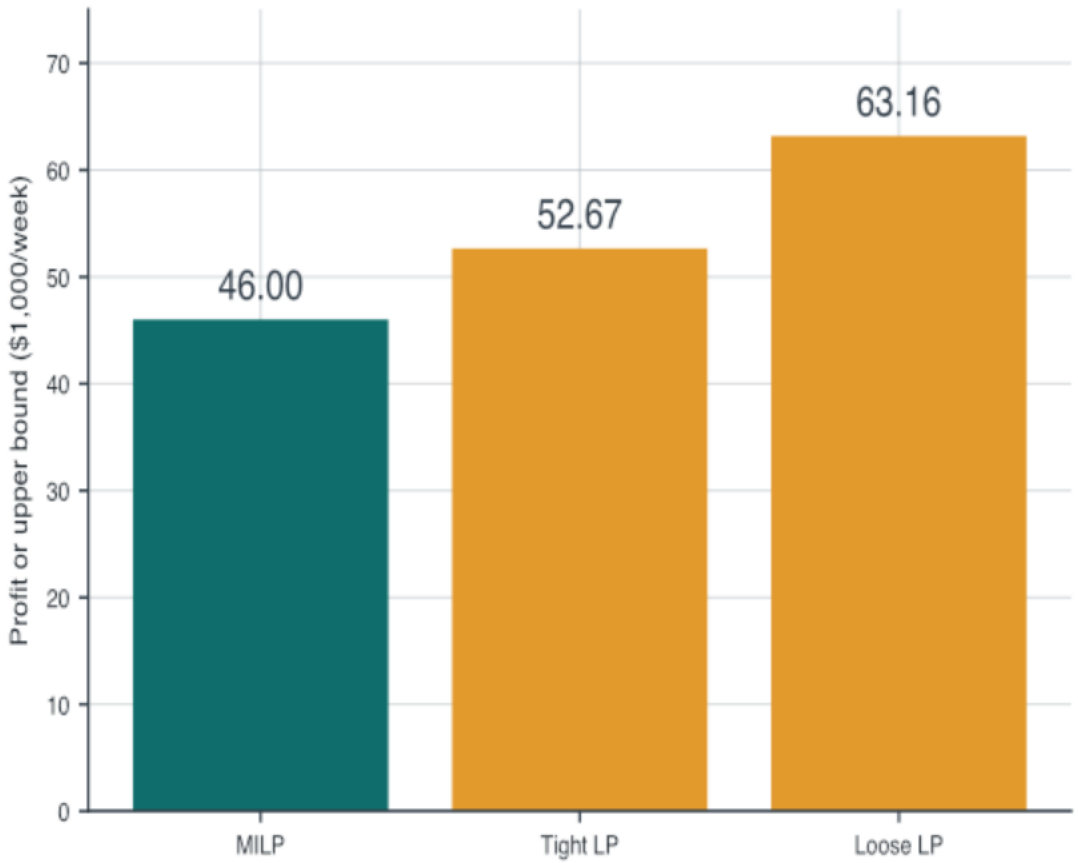


Relax the opening decisions

Allow $0 \le y_A, y_B \le 1$ temporarily.

Relaxed result: $a=2, b=8, y_A=1/3, y_B=1$.

Bound 52.667 versus integer profit 46.

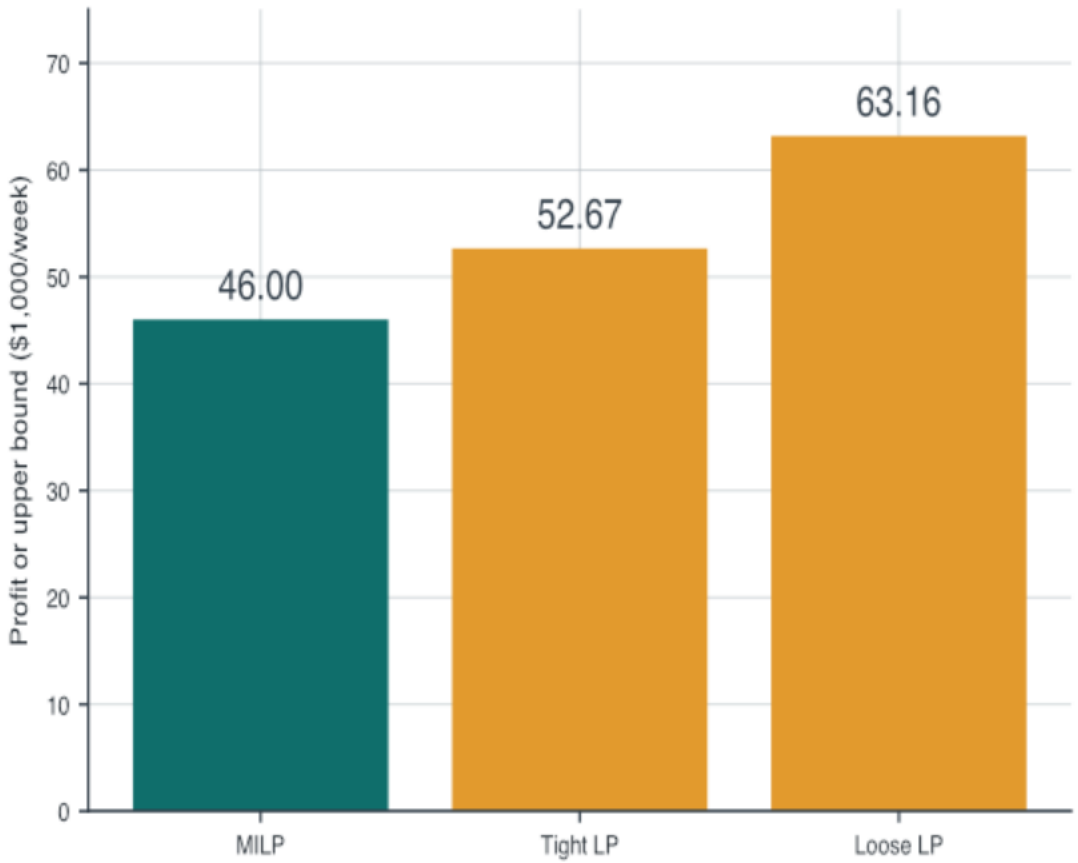


A loose bound makes the relaxation weaker

Keep $a \le 6$ and $b \le 8$ as physical limits.

Use $a \le 100yA$ and $b \le 100yB$ only for linking.

Relaxed profit becomes 63.16; the MILP stays at 46.



Branch-and-bound: maintain two values

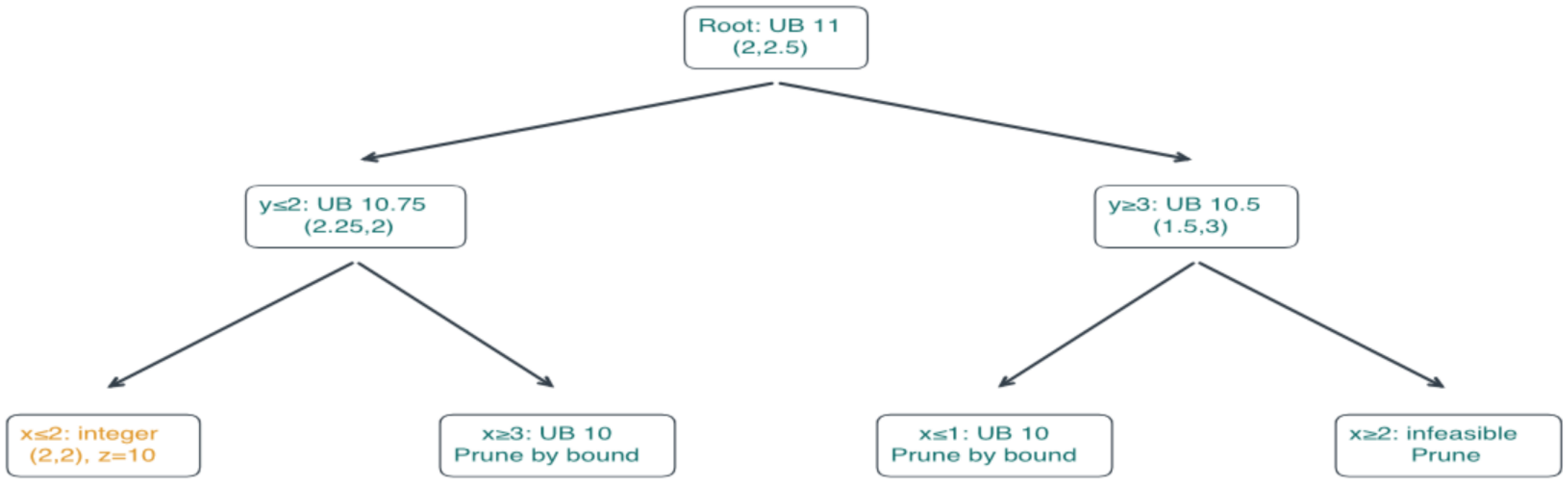
Incumbent: best feasible integer plan found so far.

Node bound: best value of a restricted LP relaxation.

For maximization, prune a node if its bound ≤ incumbent.

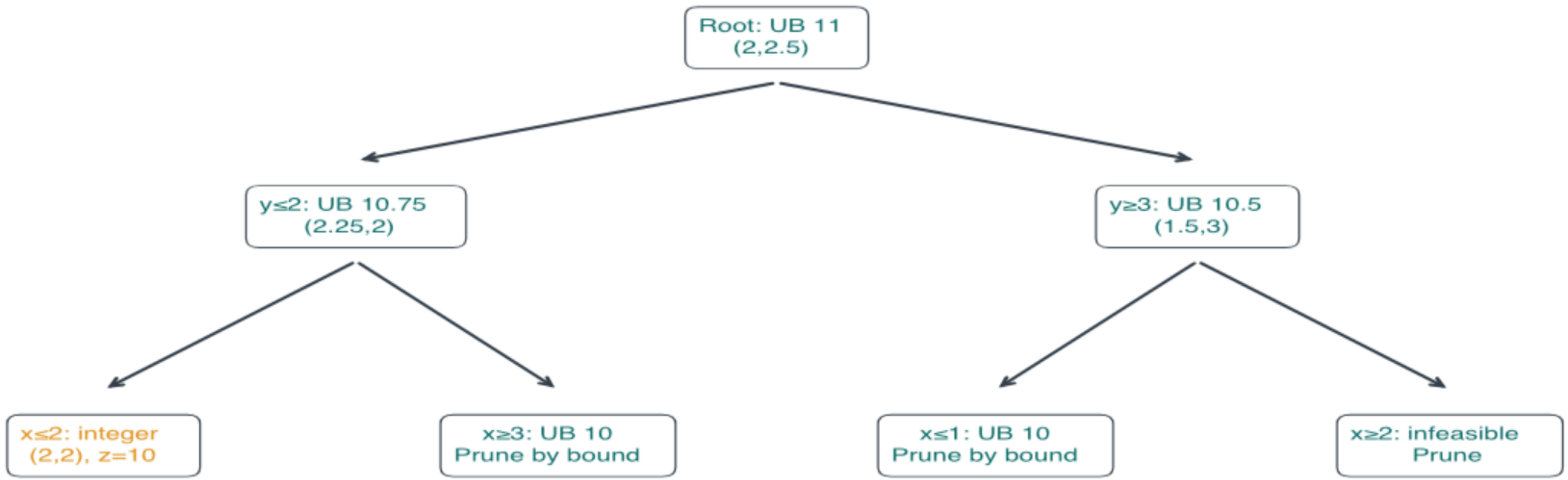
Root node: split a fractional variable

Batch LP: (x,y)=(2,2.5), upper bound 11. • Every integer y satisfies y≤2 or y≥3. • These two branches cover every integer possibility.



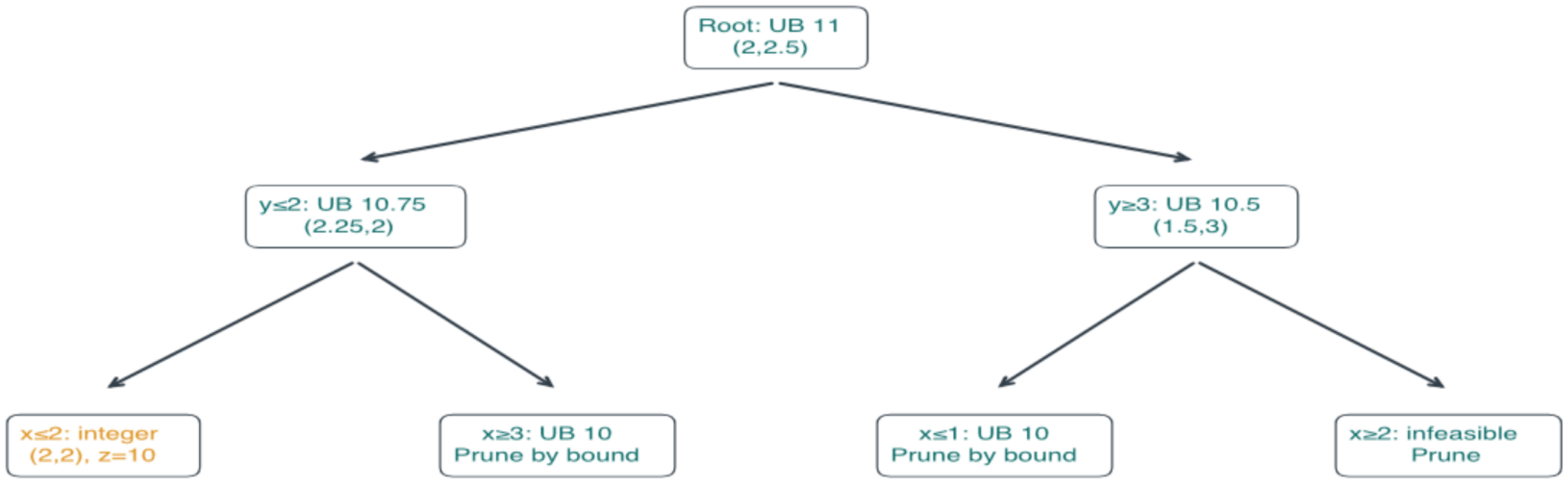
Left branch: y ≤ 2

LP result: (2.25,2), upper bound 10.75. • x is fractional, so split x≤2 or x≥3. • In the x≤2 child: (2,2), value 10, is integer.



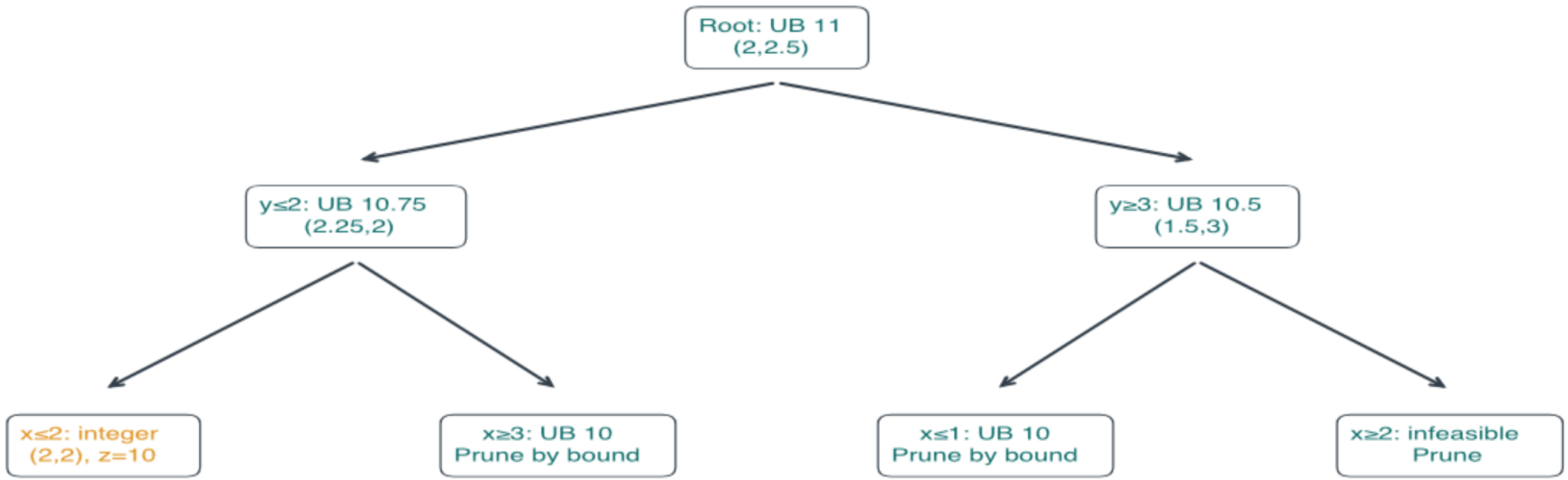
Prune the other left child

Constraints include y≤2 and x≥3. • Its LP optimum is (3,0.5), bound 10. • Bound equals incumbent 10: no better value is possible.



Right branch: y ≥ 3

LP result: (1.5,3), upper bound 10.5. • Split x≤1 or x≥2. • x≤1 gives bound 10; x≥2 is infeasible.



What certifies completion?

Feasible incumbent value: 10.

Every remaining branch is closed or bounded by 10.

Therefore the global integer optimum is 10.

Stopping early and reading the gap

A feasible plan may exist before optimality is proved.

For maximization, lower bound = incumbent value.

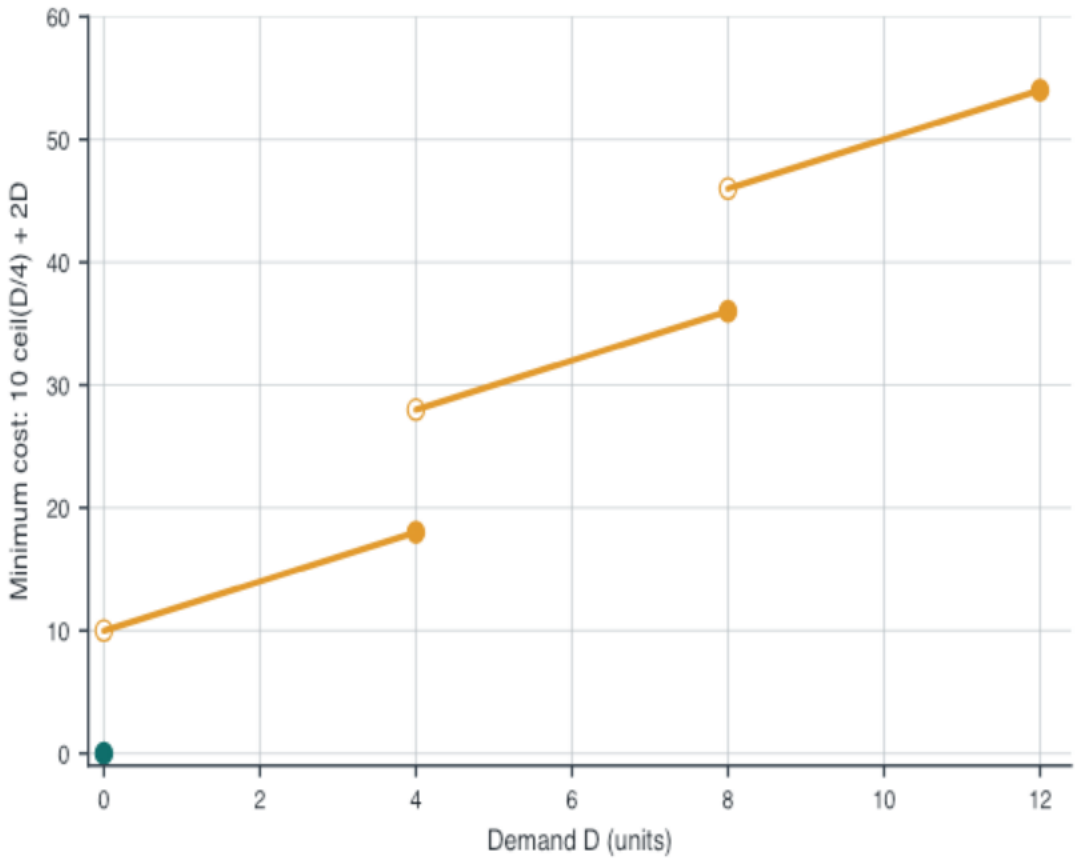
Negate a minimization dual bound to get a max upper bound.

Why LP shadow prices need care

Integer decisions can switch when data change.

Do not extrapolate an LP marginal across a MILP change.

Change the capacity or demand and re-solve the MILP.



Python lab: predict before running

Download MILP-guided-lab.ipynb.

Use Python with NumPy, SciPy, and Matplotlib.

Run cells in order; pause before each displayed answer.

Step 1: choose a fixed variable order

$v = (a, b, yA, yB)$.

Minimization coefficients: $c = (-8,-6,10,8)$.

Return to profit with profit = −result.fun.

$$A = \begin{array}{c} \begin{matrix} a & b & yA & yB \end{matrix} \\ \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & -6 & 0 \\ 0 & 1 & 0 & -8 \end{bmatrix} \end{array}$$

$$A\,v \le (12, 10, 0, 0)^T$$

Step 2: encode variable domains

integrality = [0,0,1,1].

Lower bounds = [0,0,0,0].

Upper bounds = [6,8,1,1].

Step 3: build the constraint matrix

Rows: labor, sales, link A, link B.

$A\,v \le [12,10,0,0]$.

LinearConstraint(A, $-\infty$, upper) encodes upper bounds.

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & -6 & 0 \\ 0 & 1 & 0 & -8 \end{bmatrix} \quad \begin{matrix} \\ \\ \\ \\ \end{matrix} \quad \begin{matrix} a & b & yA & yB \end{matrix}$$

$A\,v \le (12, 10, 0, 0)^T$

Step 4: solve and check status

result = milp(c, integrality=…, bounds=…, constraints=…).

Check success, status, and message.

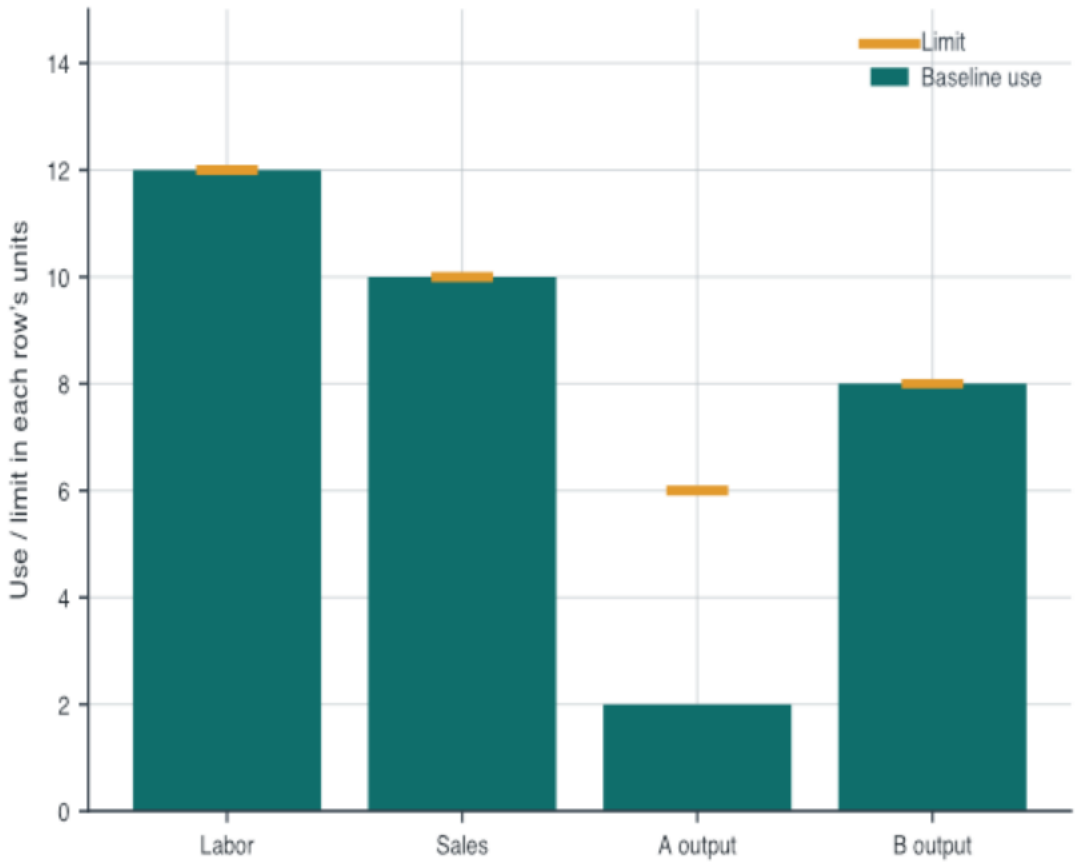
Baseline: v=(2,8,1,1), profit 46.

Step 5: audit every constraint

Compute resource_use = A @ result.x.

Check resource_use ≤ upper within a small tolerance.

Check binary values and recompute profit independently.

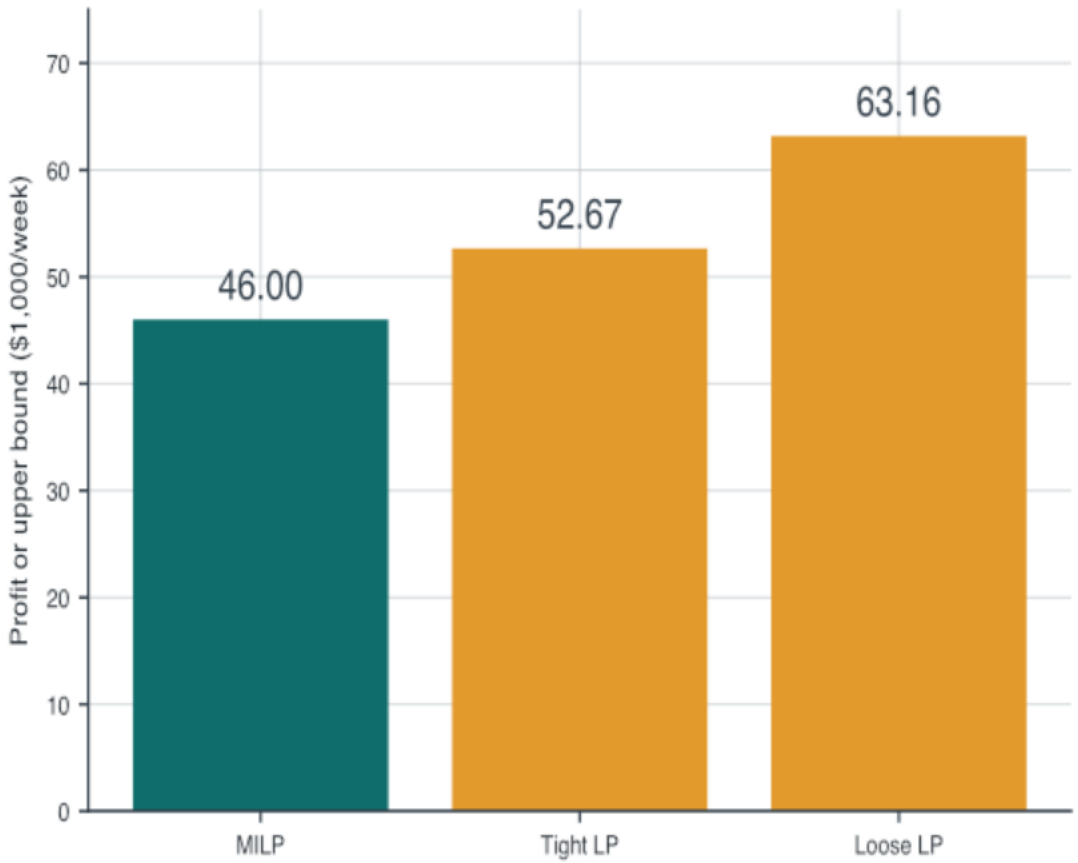


Step 6: compare relaxation and MILP

Relaxed profit: 52.667; MILP profit: 46.

Relaxed yA=1/3 is not an opening decision.

Enumerate four binary patterns to verify the MILP.

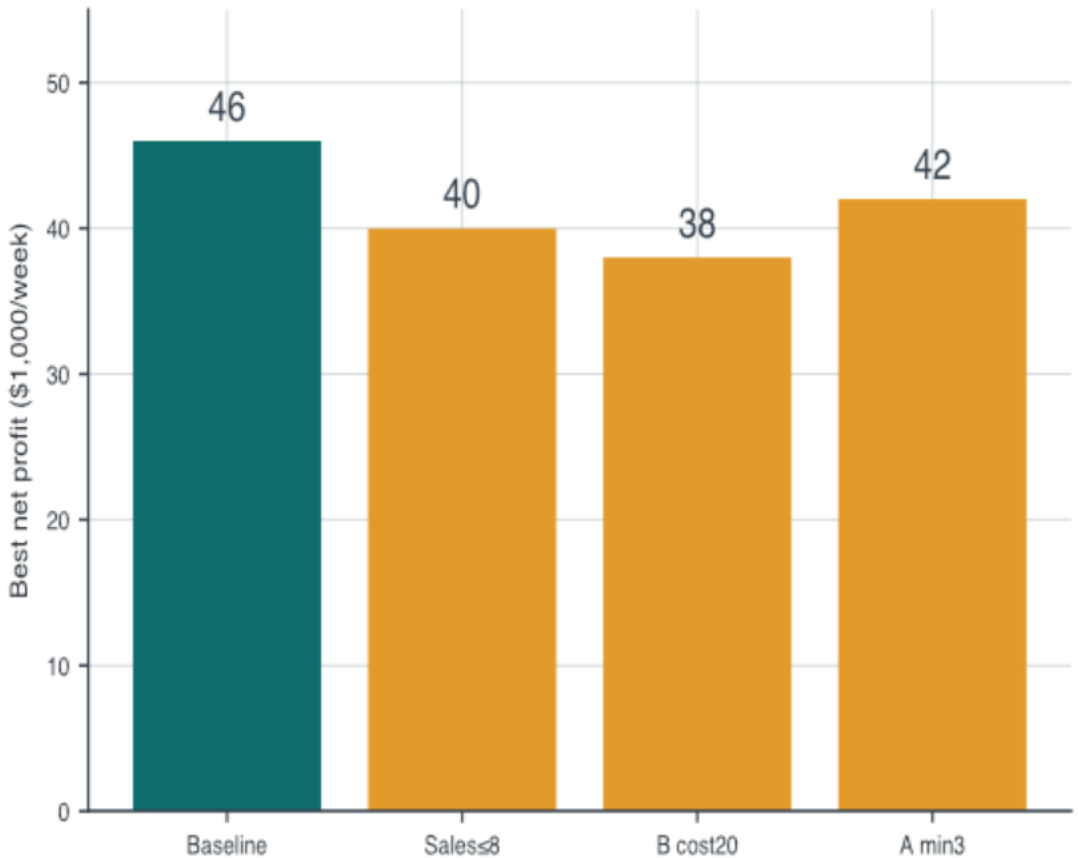


Step 7: change one assumption

Sales cap 8: open B only, profit 40.

B opening cost 20: open A only, profit 38.

A minimum batch 3: both open, a=3,b=6, profit 42.

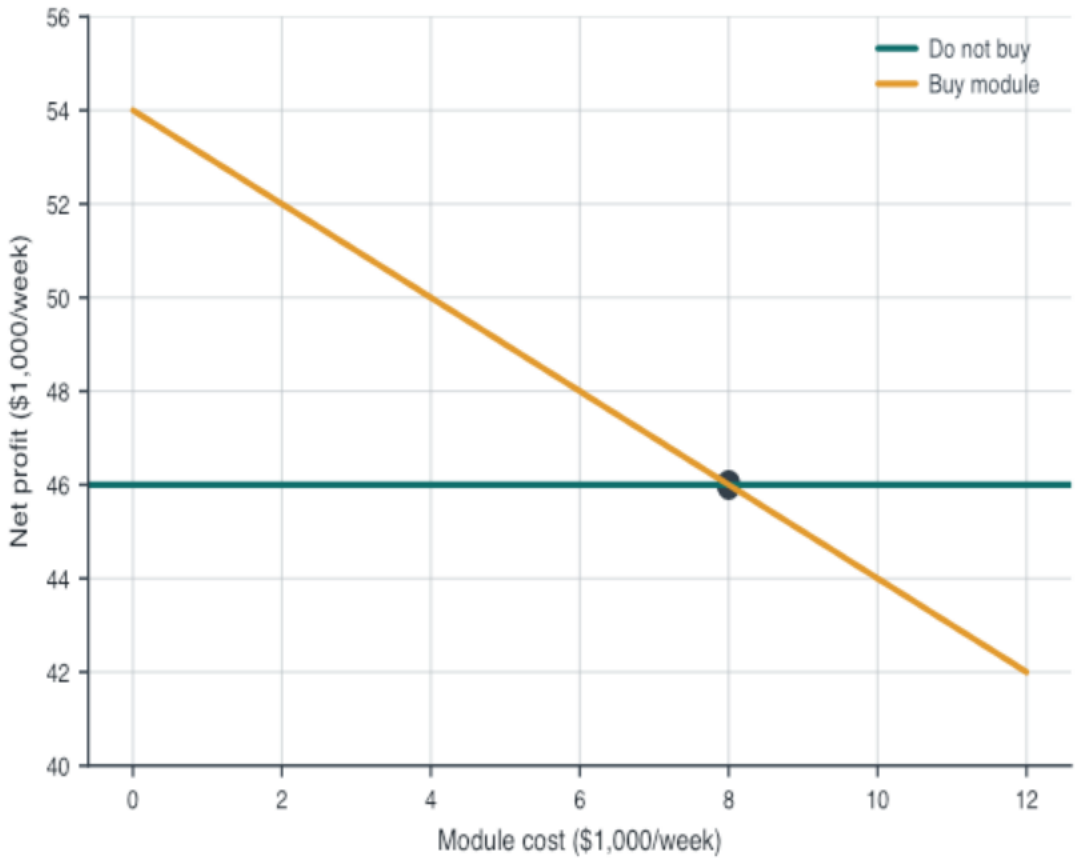


Step 8: consider an extra capacity module

$k \in \{0,1\}$; labor capacity = 12 + 4k.

Module cost = 7; subtract 7k from profit.

Optimum: k=1,a=6,b=4; net profit 47.

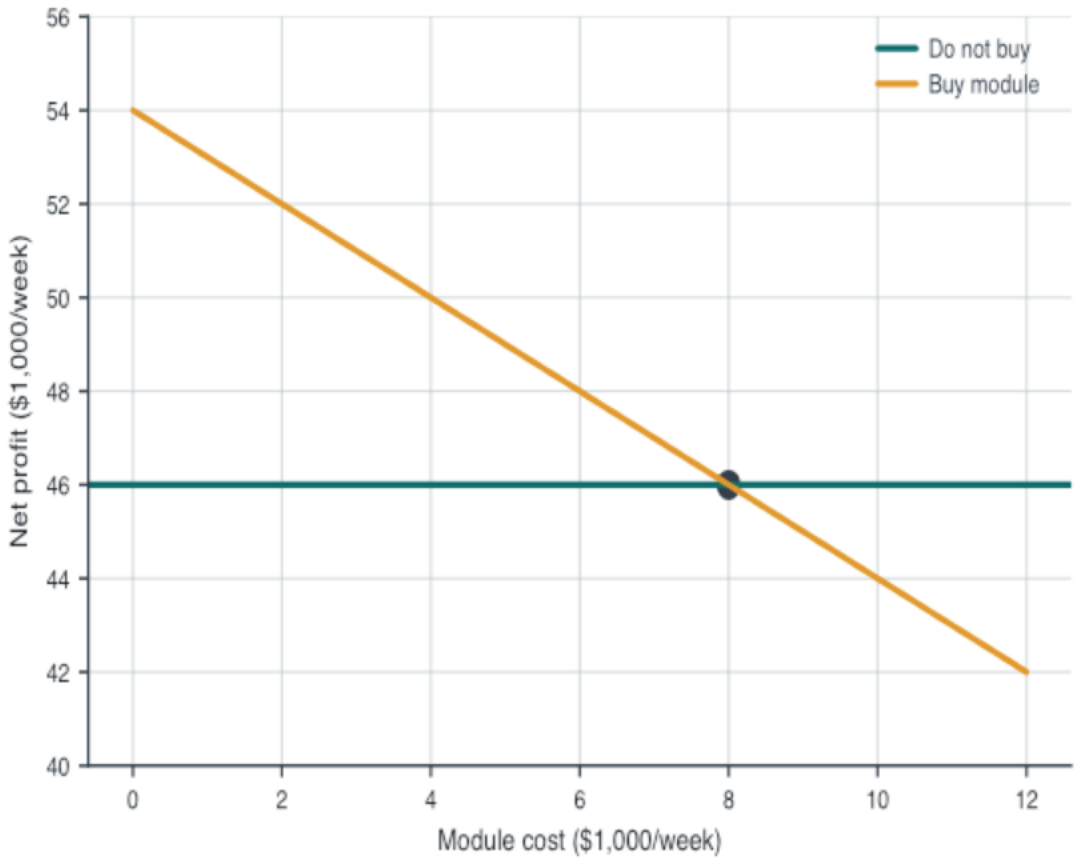


Step 9: compare purchase costs

Without purchase: profit 46.

With module before module cost: profit 54.

Cost 7 → buy; cost 10 → do not buy.



Independent exercises

1. Batch model: reduce the first resource limit to 4.

2. Plant model: reduce the sales cap to 6.

3. Add an at-most-one-line policy to the baseline.

Independent exercises: decisions and bounds

4. Module cost 8: compare buying and not buying.

5. A minimum batch 4: find the best opening pattern.

6. With incumbent 46 and max bound 48, what is certified?

Exercise solutions: batch and sales cap

1. First resource limit 4: unique best (2,2), value 10.

2. Sales cap 6: A only, a=6, profit 38.

3. At most one line: B only, b=8, profit 40.

Exercise solutions: module and minimum batch

4. Cost 8: buy or not buy both give profit 46.

5. A minimum batch 4: choose B only, profit 40.

6. $46 \le \text{optimum} \le 48$; possible improvement ≤ 2.

Exit Ticket: explain, then submit

Complete four questions on the online form.

Use your own calculations and reasoning.

Submit before opening the following solution slides.



Exit Ticket questions

1. Interpret $a \le 6y_A$ and $a \ge 3y_A$ for binary y_A.

2. Why does relaxed profit 52.667 not beat MILP profit 46?

3. Module cost 9: buy or not? Show net benefit.

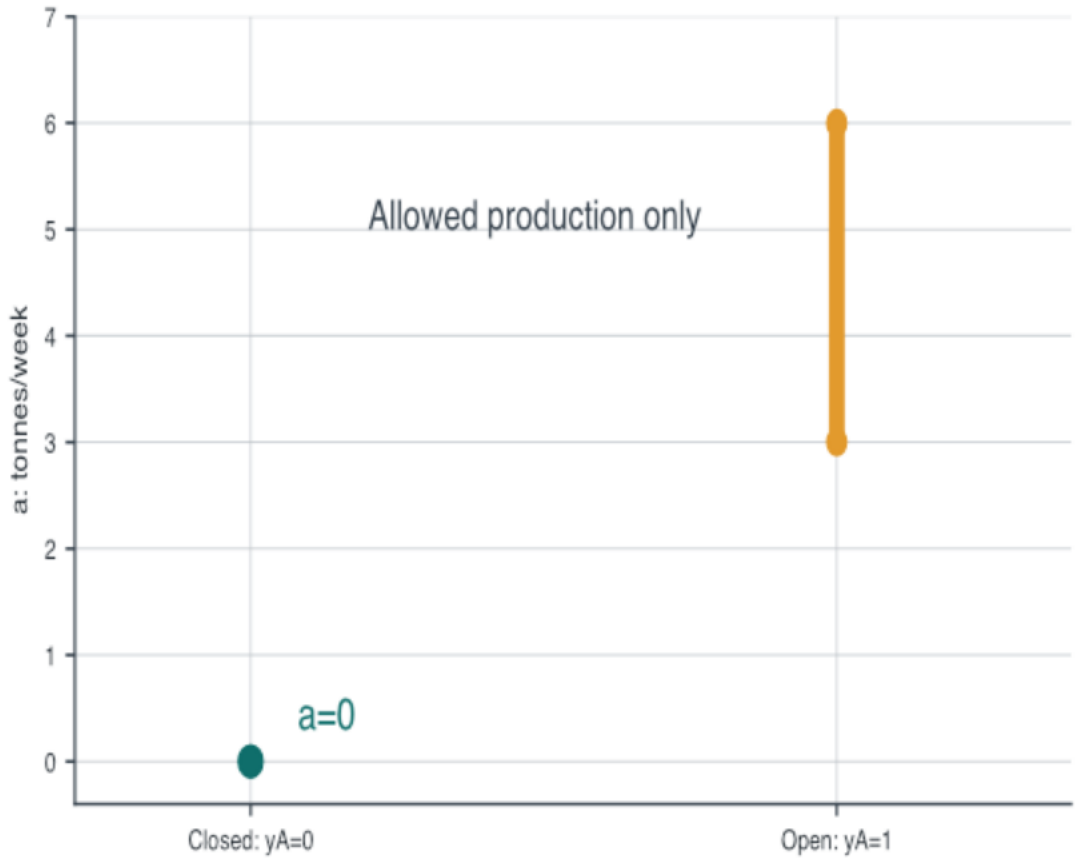
4. Incumbent 46: prune nodes with bounds 45,46,48?

Exit Ticket solutions: linking and relaxation

1. $y_A=0 \Rightarrow a=0$; $y_A=1 \Rightarrow 3 \le a \le 6$.

2. Relaxation allows $y_A=1/3$ and partial fixed cost.

52.667 is an upper bound, not an attainable MILP profit.



Exit Ticket solutions: purchase and pruning

3. Extra operating benefit $54-46=8$; cost 9: net -1.

Do not buy; best total profit remains 46.

4. Prune bounds 45 and 46; keep bound 48 for further work.

A reliable MILP workflow

Define decisions, units, domains, and valid bounds.

Solve; audit feasibility, costs, status, and gap.

Change assumptions one at a time and re-optimize.

Resources and next steps

SciPy milp: coefficients, integrality, bounds, status and gap.

Course page: videos, lab, practice, solutions and Exit Ticket.

Keep your calculations and questions for the next meeting.